Lesson4 Refraction of Light

- Facts about refraction
- Real and apparent depth
- Refractive index
- Refraction by a prism
- Dispersion
- Practical work: Refraction in glass

If you place a coin in an empty dish and move back until you *just* cannot see it, the result is surprising if someone *gently* pours in water. Try it.

Although light travels in straight lines in a transparent material, such as air, if it passes into a different material, such as water, it changes direction at the boundary between the two, i.e. it is bent. The **bending of light** when it passes from one material (called a medium) to another is called **refraction**. It causes effects such as the coin trick.

1. Facts about refraction

(i) A ray of light is bent **towards** the normal when it enters an optically denser medium at an angle, for example from air to glass as in Figure 29.1a. The angle of refraction r is less than the angle of incidence i.

(ii) A ray of light is bent **away from** the normal when it enters an optically less dense medium, for example from glass to air.

(iii) A ray emerging from a parallel-sided block is **parallel** to the ray entering, but is **displaced sideways**, like the ray in Figure 29.1a.

(iv) A ray travelling along the normal direction at a boundary is **not refracted** (Figure 29.1b).

Note 'Optically denser' means having a greater refraction effect; the actual density may or may not be greater.

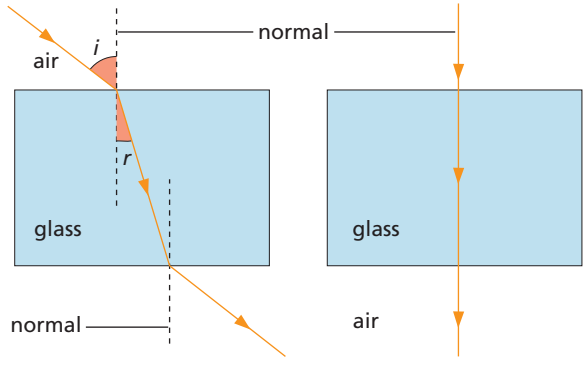


Figure 29.1 Refraction of light in glass

Practical work

Refraction in glass

Shine a ray of light at an angle on to a glass block (which has its lower face painted white or frosted), as in Figure 29.2. Draw the outline ABCD of the block on the sheet of paper under it. Mark the positions of the various rays in air and in glass.

Remove the block and draw the normals on the paper at the points where the ray enters side AB (see Figure 29.2) and where it leaves side CD.

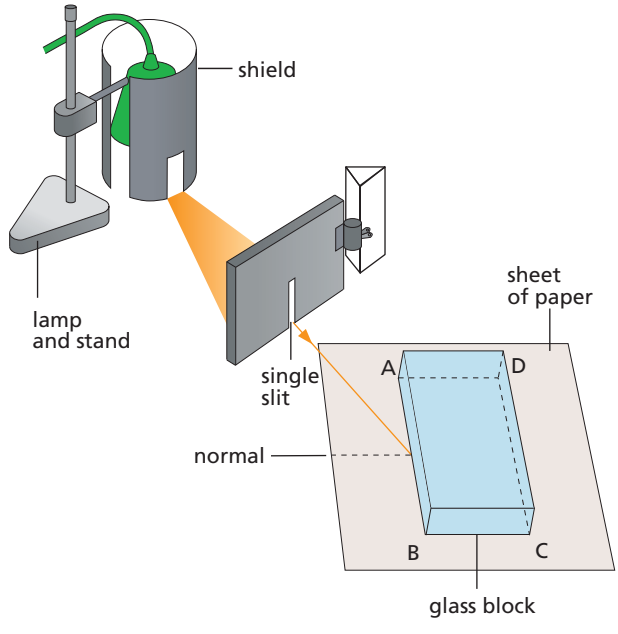


Figure 29.2

What *two* things happen to the light falling on AB? When the ray enters the glass at AB, is it bent towards or away from the part of the normal in the block? How is it bent at CD? What can you say about the direction of the ray falling on AB and the direction of the ray leaving CD?

What happens if the ray hits AB at right angles?

2. Real and apparent Depth

Rays of light from a point O on the bottom of a pool are refracted away from the normal at the water surface because they are passing into an optically less dense medium, i.e. air (Figure 29.3). On entering the eye they appear to come from a point I that is *above* O; I is the virtual image of O formed by refraction. The apparent depth of the pool is less than its real depth.

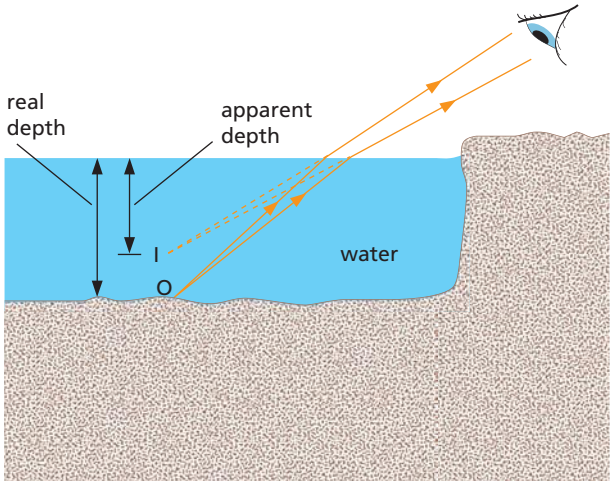


Figure 29.3 A pool of water appears shallower than it is.



Figure 29.4 A pencil placed in water seems to bend at the water surface. Why?

3. Refractive index

Light is refracted because its speed changes when it enters another medium. An analogy helps to explain why.

Suppose three people A, B, C are marching in line, with hands linked, on a good road surface. If they approach marshy ground at an angle (Figure 29.5a), person A is slowed down first, followed by B and then C. This causes the whole line to swing round and change its direction of motion.

In air (and a vacuum) light travels at 300 000 km/s (3×10^8 m/s); in glass its speed falls to 200 000 km/s (2×10^8 m/s) (Figure 29.5b). The **refractive index**, n, of a medium, in this case glass, is defined by the equation

$$\text{refractive index, } n = \frac{\text{speed of light in air (or a vacuum)}}{\text{speed of light in medium}}$$

$$\text{for glass, } n = \frac{300\,000 \text{ km/s}}{200\,000 \text{ km/s}} = \frac{3}{2}$$

Experiments also show that

$$n = \frac{\text{sine of angle between ray in air and normal}}{\text{sine of angle between ray in glass and normal}}$$

$$= \frac{\sin i}{\sin r} \quad \text{(see Figure 29.1a)}$$

The more light is slowed down when it enters a medium from air, the greater is the refractive index of the medium and the more the light is bent.

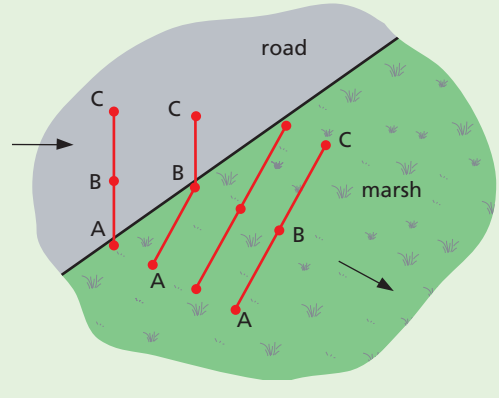


Figure 29.5a

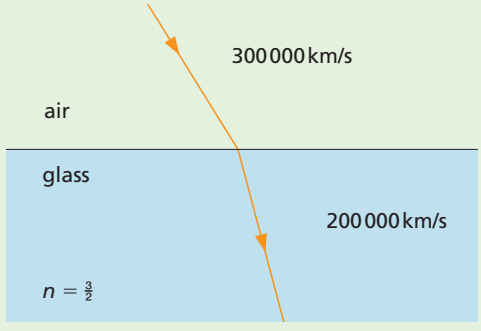


Figure 29.5b

We saw earlier (Chapter 25) that water waves are refracted when their speed changes. The change in the direction of travel of a light ray when its speed changes on entering another medium suggests that light may also be a type of wave motion.

4. Refraction by a prism

In a triangular glass prism (Figure 29.6a), the bending of a ray due to refraction at the first surface is added to the bending of the ray at the second surface (Figure 29.6b); the overall change in direction of the ray is called the **deviation**.

The bendings of the ray do not cancel out as they do in a parallel-sided block where the emergent ray, although displaced, is parallel to the incident ray.

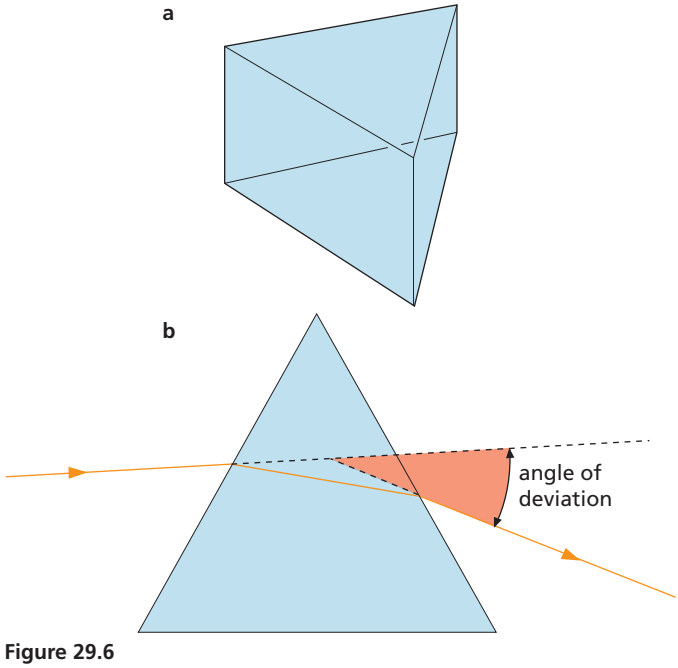


Figure 29.6

5. Dispersion

When sunlight (white light) falls on a triangular glass prism (Figure 29.7a), a band of colours called a **spectrum** is obtained (Figure 29.7b). The effect is termed **dispersion**. It arises because white light is a mixture of many colours; the prism separates the colours because the refractive index of glass is different for each colour (it is greatest for violet light).

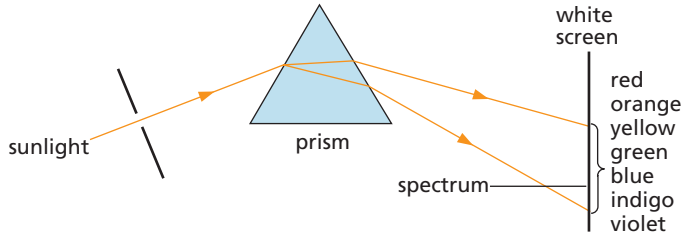


Figure 29.7a Forming a spectrum with a prism

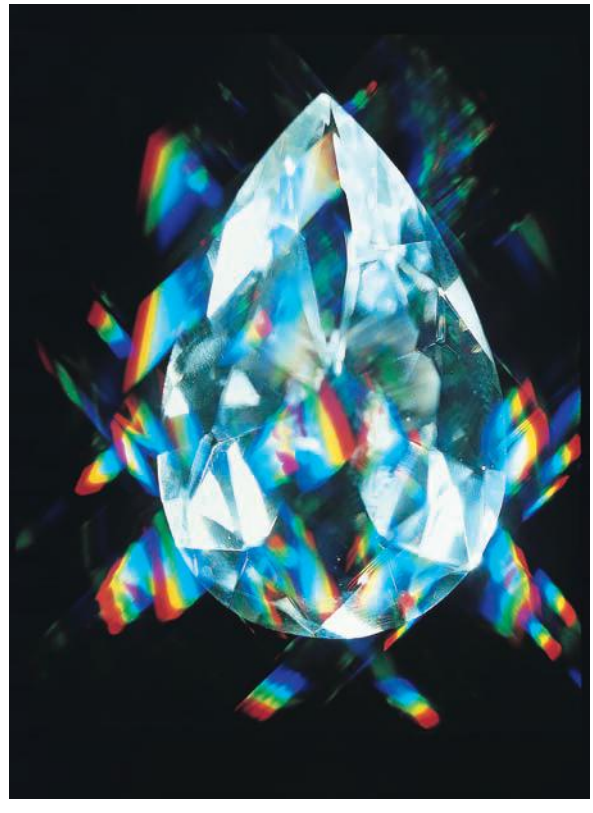


Figure 29.7b White light shining through cut crystal can produce several spectra.

Questions

1 Figure 29.8 shows a ray of light entering a rectangular block of glass.
- **a** Copy the diagram and draw the normal at the point of entry.
- **b** Sketch the approximate path of the ray through the block and out of the other side.

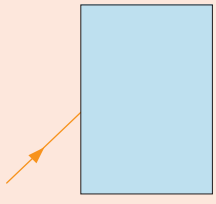


Figure 29.8

2 Draw two rays from a point on a fish in a stream to show where someone on the bank will see the fish. Where must the person aim to spear the fish?

3 What is the speed of light in a medium of refractive index 6/5 if its speed in air is 300 000 km/s?

4 Figure 29.9 shows a ray of light OP striking a glass prism and then passing through it. Which of the rays A to D is the correct representation of the emerging ray?

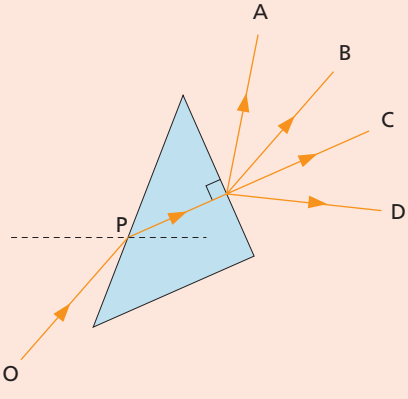


Figure 29.9

5 A beam of white light strikes the face of a prism. Copy Figure 29.10 and draw the path taken by red and blue rays of light as they pass through the prism and on to the screen AB.

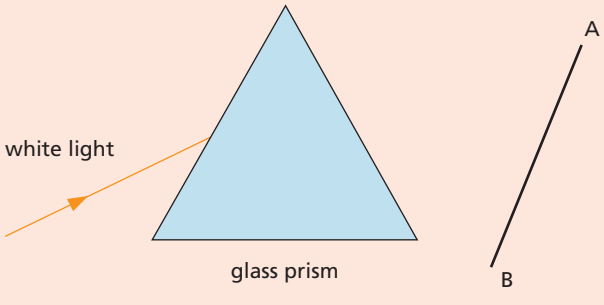


Figure 29.10

6 Which diagram in Figure 29.11 shows the ray of light refracted correctly?

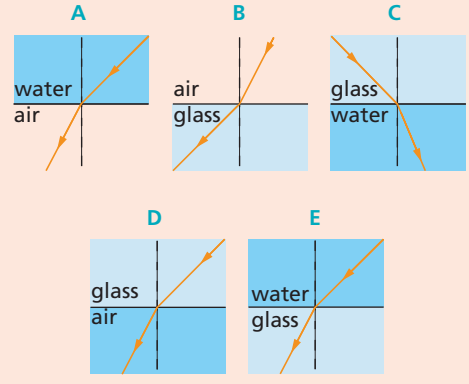


Figure 29.11

7 Which diagram in Figure 29.12 shows the correct path of the ray through the prism?

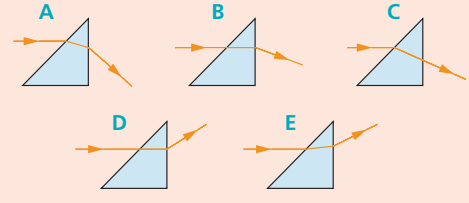


Figure 29.12

Checklist

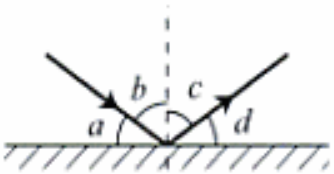
After studying this chapter you should be able to

- state what the term refraction means,
- give examples of effects that show light can be refracted,
- describe an experiment to study refraction,
- draw diagrams of the passage of light rays through rectangular blocks and recall that lateral displacement occurs for a parallel-sided block,
- recall that light is refracted because it changes speed when it enters another medium,
- recall the definition of refractive index as n = speed in air/speed in medium,
- recall and use the equation $n = \sin i / \sin r$
- draw a diagram for the passage of a light ray through a prism,
- explain the terms spectrum and dispersion,
- describe how a prism is used to produce a spectrum from white light.

Practice Exercises

1-State the law of reflection

2-The figure below shows a light ray reflecting off a smooth surface.



A-The angle of incidence is …….

B-The angle of reflection is …….

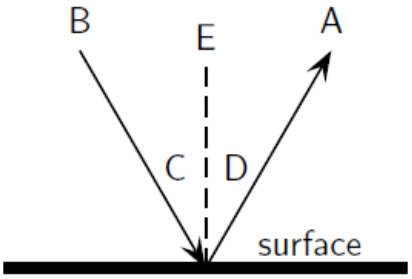
C-If a=42^0, what is the angle of reflection?

3-A light ray strikes a mirror at an angle of 72^0 to the normal line.

A-What is the angle of reflection?

B-What is the angle between the incident ray and reflected ray?

4-a ray strikes a mirror at angle of 26^0 to the mirror surface. What is the angle between the incident ray and the reflected ray?



5-An incident ray strikes a smooth reflective surface at an angle of 33° to the surface normal. What is the angle of reflection?

……
……..

6-In the diagram, label the following:

(a) normal ……

(b) angle of incidence ……..

(c) angle of reflection ……..

(d) incident ray …….

(e) reflected ray ……..

7- A ray of light strikes a surface at 15° to the surface normal. Draw a ray diagram showing the incident ray, reflected ray and surface normal. Calculate the angles of incidence and reflection and fill them in on your diagram.

8-Calculate the speed of light in water, glass, and diamond using the index of refraction and the speed of light in a vacuum (3×10^8m/s)

9. A ray of light traveling from air into crown glass strikes the surface at an angle of 30 degrees. What will the angle of refraction be?

10. Light travels through a liquid at 140,000 miles/second. What is the index of the liquid?

11. Light traveling through air encounters a second medium which slows the light to 100,000 miles/second. What is the index of the second medium?

12. What is the index of refraction of a refractive medium if the angle of incidence in air is 30 degrees and the angle of refraction is 15 degrees?

13. What is the index of refraction of a refractive medium if the angle of incidence in air is 40 degrees and the angle of refraction is 29 degrees?

14. What is the index of refraction of a liquid if the angle of incidence in air is 35 degrees and the angle of refraction is 14 degrees?

15. If the angle of incidence of light traveling through air, striking water, is 30 degrees, what is the angle of refraction?

16. If the index of refraction for a certain glass is 1.50, and the angle of refraction is 15 degrees for a ray of light traveling from air, what is the angle of incidence?

17. What is the velocity of light in miles per second in a material with an index of 2?

18.Light moving through air is incident on a piece of crown glass at an angle of $45°$. What is the angle of refraction?

